Arrangements 2

1. Find the number of permutations of the following words if all the letters are used:
   a. BOB
   b. ALAN
   c. SNEEZE
   d. TASMANIA
   e. BEGINNER
   f. FOOTBALLS
   g. EQUILATERAL
   h. COMMITTEE
   i. WOOLLOOMOOLOO
2. The six digits $1, 1, 1, 2, 2, 3$ are used to form a six-digit number. How many numbers can be formed?
3. Six coins are lined up on a table. Find how many patterns are possible if there are:
   a. Five tails and one head
   b. Four heads and two tails
   c. Three tails and three heads
4. Eight balls, identical except for colour, are arranged in a line. Fin dhow many different arrangements are possible if:
   a. All balls are of a different colour
   b. There are seven red balls and one white ball
   c. There are six red balls, one white ball and one black ball
   d. There are three red balls, three white balls and two black balls.
5. Five identical green chairs and three identical red chairs are arranged in a row. Find how many arrangements are possible:
   a. If there are no restrictions
   b. If there must be a green chair on either end.
6. A motorist travels through eight sets of traffic lights, each of which is red or green. He is forced to stop at three sets of lights.
   a. In how many ways could this happen?
   b. What other number of red lights would give an identical answer to part (a)?
7. Find how many ways the letters of the word SOCKS can be arranged in a line:
   a. Without restriction
   b. So that the two Ss are together
   c. So that the two Ss are separated by at least one other letter
   d. So that the K is somewhere to the left of the C.
8. a. Find the number of arrangements of the letters in SLOOPS if:
      i. there are no restrictions
      ii. the two Os are together
      iii. the two Os are to be separated
      iv. the Os are together and the Ss are together

   b. How many arrangements of the letters in TATTOO are there, if the two Os are to be separated?
9. Find how many ways the letters of the word DECISIONS can be arranged:
   a. Without restriction
   b. So that the vowels and consonants alternate
   c. So that the vowels come first followed by the consonants
   d. So that the N is somewhere to the right of the D.
10. In how many ways can the letters of the work PROPORTIONALITY be arranged so that the vowels and consonants still occupy the same places?
11. A form has ten questions in order, each of which requires the answer 'Yes' or 'No'. Find the number of ways the form can be filled in:
    i. Without restriction
    ii. If the first and last answers are 'Yes'

iii. If two are 'Yes' and eight are 'No'
iv. If five are 'Yes' and five are 'No'
v. If more than seven answers are 'Yes'
vi. If an odd number of answers are 'Yes'
vii. If exactly three answers are 'Yes', and they occur together
viii. If the first and last answers are 'Yes' and exactly four more are 'Yes'

12. Containers are identified by a row of coloured dots on their lids, the colours used are yellow, green and purple. In any arrangement, there are to be no more than three yellow dots, so more than two green dots are no more than one purple dot.
    a. If six dots are used, what is the number of possible codes?
    b. What is the number of different codes possible if only five dots are used?
13. a. How many five-letter words can be formed by using the letters of the word STRESS?
    b. How many five-letter words can be formed by using the letters of the word BANANA?
14. Find how many arrangements of the letters of the word TRANSITION are possible if:
    a. There are no restrictions
    b. The Is are together
    c. The Is, Ns and Ts are together
    d. The Ns occupy the end positions
    e. An N occupies the first but not the last position
    f. The letter N is not at either end
15. Bob is about to hang his eight shirts in the wardrobe. He has four different styles of shirt, two identical ones of each particular style. How many different arrangements are possible if no two identical shirts are next to one another?
16. Find how many arrangements there are to the letters of the word GUMTREE if:
    a. There are no restrictions
    b. The Es are together
    c. The Es are separated by:
        i. One letter
        ii. Two letters
        iii. Three letters
        iv. Four letters
        v. Five letters
    d. The G is somewhere between the two Es
    e. The M is somewhere to the left of both Es and the U is somewhere between them
    f. The G is somewhere to the left of the U and the M is somewhere to the right of the U

Answer:

1. a. 3; b. 12; c. 120; d. 6720; e. 10080; f. 90720; g. 4,989,600; h. 45,460; i. 25,740
2. 60
3. a. 6; b. 15; c. 20
4. a. 40,320; b. 8; c. 56; d. 560
5. a. 56; b. 20
6. a. 56; b. 5
7. a. 60; b. 24; c. 36; d. 30 (half of them)
8. a. i. 180, ii. 60, iii. 120, iv, 24; b. 40
9. a. 90,720; b. 720; c. 720; d. 45,360 (half of them)
10. 2,721,600
11. a. 1024; b. 256; c. 45; d. 252; e. 56; f. 512; g. 8; h. 70
12. a. 60; b. 60
13. a. 120; b. 60
14. a. 453,600; b. 90,720; c. 5040; d. 10,080; e. 80,640; f. 282,240
15. 864
16. a. 2520; b. 720; c. i. 600, ii. 480, iii. 360, iv. 240, v. 120; d. 840; e. 210; f. 420